imension (Aspect)	ID	Competence Name	Key Simplified Focus (Applied to all 12 Sectors)	Blue Biotech	Coastal Tourism	Desalination	nfra & Robotics	Living Res.	Non-living Res	enewable Ener	aritime Defen	aritime Transp	Port Activities	R&I	Ship Repair
A. Understanding the Blue Economy (Skills)	—	*Technical/practical abilities*	Policy use & marine science basics,	X	X	X	X	X	X	X	X	X	X	X	X
A. Understanding the Blue Economy (Competences)	A.1	Ocean literacy	Communicating ocean value and agency	X	X	X	X	X	X	X	X	X	X	X	X
	A.2	Blue systems thinking	Mapping planetery system connections	X	X	X	X	X	X	X	X	X	X	X	X
	A.3	Blue economy regulations	Following blue values laws, inclusivity, justice an	X	X	X	X	X	X	X	X	X	X	X	X
B. Digital & Data Skills (Skills)	—	*Technical/practical abilities*	Software use & cyber defense,	X	X	X	X	X	X	X	X	X	X	X	X
B. Digital & Data Skills (Competences)	B.1	Data & digital proficiency	Using digital tools (GIS) & data in-research	X	X	X	X	X	X	X	X	X	X	X	X
	B.2	Digital communication	Online collaboration & outreach, counter-mapin	X	X	X	X	X	X	X	X	X	X	X	X
	B.3	Cybersecurity & maritime safety	Protecting digital systems,	X	X	X	X	X	X	X	X	X	X	X	X
	B.4	Open science & data sharing	Transparent data access, civic repositories, indig	X	X	X	X	X	X	X	X	X	X	X	X
C. Sustainability & Resilience (Skills)	—	*Technical/practical abilities*	Green design & resource planning,	X	X	X	X	X	X	X	X	X	X	X	X
C. Sustainability & Resilience (Competences)	C.1	Sustainable resource management	Responsible resource use, zerocarbon, zeroplast	X	X	X	X	X	X	X	X	X	X	X	X
	C.2	Circular economy principles	Reducing waste & recycling, water retention	X	X	X	X	X	X	X	X	X	X	X	X
	C.3	Climate adaptation & coastal resilience	Building coastal buffers, off-shore and deep sea	X	X	X	X	X	X	X	X	X	X	X	X
	C.4	Ecosystem-based management (EBM)	Holistic nature-based care, blue governance	X	X	X	X	X	X	X	X	X	X	X	X
D. Business Innovation & Governance (Skills)	—	*Technical/practical abilities*	Value mapping & financial tracking,	X	X	X	X	X	X	X	X	X	X	X	X
D. Business Innovation & Governance (Competences)	D.1	Value chain thinking	Optimizing production flows, non-human and mo	X	X	X	X	X	X	X	X	X	X	X	X
	D.2	Leadership & blue innovation	Leading sustainable change, blue growth, blue ju	X	X	X	X	X	X	X	X	X	X	X	X
	D.3	Blue finance & investment	Finding sustainable funding, blue finance, blue e	X	X	X	X	X	X	X	X	X	X	X	X
	D.4	Ethical & participatory governance	Inclusive decision-making, blue governance, blu	X	X	X	X	X	X	X	X	X	X	X	X

The "Sociological Spine" of Clusters (see: Blue Clusters for Microcredentials)
While maritime academies and polytechnics focus on the Hardware of the Blue Economy (the what and how of technology), the University of Szczecin provides the Software (the who and why of society).
Without this software, the hardware fails. A wind farm cannot be built if the local community blocks the permit. A biotechnology lab cannot function if it violates ethical treaties on genetic resources. The "Sociological Spine" runs through every cluster, ensuring that technical projects are viable, legal, and socially accepted.

Cluster 1: Living Resources & Biotechnology
The "Bio-Economy"
The Technical View (Hardware): "How do we extract active compounds from marine ecosystem?"
The Sociological Spine (Software - Layer D.4): "Who owns, who govern who has agency?"(who owns genes?)
The Problem: Technical competence alone leads to "biopiracy"—companies extracting genetic resources from the Global South or local waters without sharing benefits, leading to lawsuits and blocked trade.
Szczecin's Contribution: We teach the Nagoya Protocol and Benefit-Sharing Ethics. We train the "Blue Diplomat" who ensures that a biotech startup doesn't just patent a gene, but creates a fair contract with the provider country or community.
Micro-credential Value: A "Bio-value Communication" credential teaches a scientist how to explain to a local fisherman why preserving a sponge species is more valuable than trawling over it.

Cluster 2: Energy, Resources & Desalination - Maritime Spatial Planning (MSP)
The "Hard Economy"
The Technical View (Hardware): "How do we anchor a 15MW turbine in the Baltic seabed?"
The Sociological Spine (Software - Layer D.4): "Who owns the horizon?" (offshore conflict resolution)
The Problem: Engineers see empty space; sociologists see "lived space." Fishermen see their heritage; coastal residents see a ruined view. This mismatch causes the #1 bottleneck in green energy: Social Rejection (NIMBYism).
Szczecin's Contribution: We teach Conflict Mediation and Participatory Maritime Spatial Planning (MSP). We don't just draw zones on a map; we train the facilitators who run the town halls, ensuring that wind farms coexist with fishing grounds.
Micro-credential Value: A "Multi-use Coordination" credential allows a project manager to design a wind farm that legally and socially integrates aquaculture, turning opponents into partners.

Cluster 3: Transport, Ports & Infrastructure
The "Blue Spine"
The Technical View (Hardware): "How do we optimize container throughput using AI?"
The Sociological Spine (Software - Layer D.4): "Who belongs to the port city?"
The Problem: Modern ports are fenced-off, automated zones. This leads to "Demaritimisation"—the city turns its back on the port. Locals see only pollution and traffic, not pride or jobs. This threatens the port's "Social License to Operate."
Szczecin's Contribution: We teach Re-Maritimisation. We train urban planners and port officers to reintegrate the port into the city's cultural identity—opening waterfronts, creating "Port Centers," and making the port a source of civic pride, not just profit.
Micro-credential Value: A "Port-City Interface Governance" credential teaches a port authority how to build public trust, reducing friction for expansion projects.

Cluster 4: Tourism, Defence & Robotics
The "Service & Security" Layer
The Technical View (Hardware): "How do we deploy autonomous drones for surveillance?"
The Sociological Spine (Software - Layer D.4): "Who are we protecting?"
The Problem: Surveillance can feel like militarization. Tourism can feel like exploitation ("Disneyfication" of the coast).
Szczecin's Contribution: We teach Ocean Literacy and Blue Citizenship. We transform "Security" from a military concept into a social one—protecting the common heritage. We transform "Tourism" into "Stewardship," where visitors pay to restore the ecosystem.
Micro-credential Value: An "Ocean Literacy & Stewardship" credential trains a tourism operator to be an educator, turning a simple boat tour into a citizen-science expedition that gathers data and funding for conservation.
Summary: The "Double-License" Graduate
A graduate from a technical academy has a Technical License (they know how to do it).
A graduate with University of Szczecin micro-credentials has a Social License (they know how to get it approved).

We do not compete with engineering schools. We complete them. We provide the essential governance, ethics, and social skills that prevent the Blue Economy from stalling due to resistance, lawsuits, and cultural blindness.

This is the "Sociological Spine" that the University of Szczecin adds to the technical curriculum.

The "Blue Values" Definition

Before entering the matrix, these are the core values derived from the texts that guide the sociological interventions:

Intergenerational Equity: Protecting the ocean for future "People of the Sea."

Procedural Justice: Fair inclusion in decision-making (not just consultation).

Solidarity: The port/coast as a shared community asset, not just a corporate zone.

Stewardship: Moving from "extracting resources" to "caretaking ecosystems."

Trans-corporeality: Acknowledging that human health and ocean health are biologically linked (Marinisation).

Cluster 1: Living Resources & Biotechnology (The Bio-Economy)

Focus: From Extraction to Stewardship

BlueComp Dimension	Technical Skill (Existing)	Sociological Concept / Theory (Szczecin Add)	Blue Value & Justification
A.2 Systemic Thinking	Stock-prey relationships	Theory: Marinisation (Physiological) Understanding how marine bio-compounds affect humar	Value: Trans-corporeality Justification: Students must understand we are not sepa
D.4 Ethics & Governance	Nagoya Protocol compliance	Concept: Benefit Sharing / Blue Justice Ensuring local communities profit from genetic data extr	Value: Equity Justification: Prevents "Biopiracy." Technical access to g
A.1 Ocean Literacy	Sustainable seafood info	Theory: Occupational Identification Transitioning the "Fisher" identity to "Ocean Farmer/Guar	Value: Dignity Justification: Fishers often resist regulation because it th

Cluster 2: Energy, Resources & Desalination (The Hard Economy)

Focus: Space, Conflict, and Co-existence

BlueComp Dimension	Technical Skill (Existing)	Sociological Concept / Theory (Szczecin Add)	Blue Value & Justification
A.3 Regulations (MSP)	Zoning & De-confliction	Theory: Maritime Territorial Community Treating the sea as a "lived space" (Territory) rather than	Value: Recognition Justification: Engineers see coordinates; locals see herit

C.4 Ecosystem Management	Environmental Impact Assessm	Theory: The Marine Factor (Agency) Recognizing the ocean's capacity to destroy infrastructur	Value: Precautionary Principle Justification: Deep-sea mining risks are unknown. Sociol
D.4 Participatory Gov	Stakeholder Consultation	Concept: Social License to Operate (SLO) Moving from "Permitting" (Legal) to "Acceptance" (Social	Value: Procedural Justice Justification: A legal permit does not stop sabotage or lit
Cluster 3: Transport, Ports & Infrastructure (The Blue Spine)			
Focus: Re-connecting the City and the Sea			
BlueComp Dimension	Technical Skill (Existing)	Sociological Concept / Theory (Szczecin Add)	Blue Value & Justification
A.2 Systemic Thinking	Port-Hinterland Logistics	Theory: De-maritimisation Diagnosing the cultural/physical separation of the city fr	Value: Cultural Heritage Justification: A port that is culturally invisible to its city lo
D.3 Stakeholder Engagement	Port Community Systems (PCS)	Concept: Re-maritimisation Active policies to reintegrate port culture into civic life (e	Value: Solidarity Justification: Rebuilding the "Maritime Community" ensu
C.2 Circular Economy	Ship Recycling / Waste	Concept: Industrial Ecology as Social Contract Ports as hubs for circular living, not just transit.	Value: Responsibility Justification: The port takes responsibility for the lifecycl
Cluster 4: Tourism, Defence & Robotics (Service & Security)			
Focus: Literacy, Citizenship, and Peace			
BlueComp Dimension	Technical Skill (Existing)	Sociological Concept / Theory (Szczecin Add)	Blue Value & Justification
A.1 Ocean Literacy	VR/AR Experiences	Concept: Blue Citizenship Transforming tourists from "consumers of views" to "citiz	Value: Stewardship Justification: Tourism is extractive unless it builds a "Ste
D.4 Governance	Civil-Military Synergy	Theory: Trans-Oceanisation Planetary governance where the ocean is a single, shared	Value: Peace & Cooperation Justification: Security is no longer just national; it is plan
B.3 Cybersecurity	Data Protection	Concept: The Oceanic Turn (Sociology) Understanding data not as numbers, but as the "Voice of	Value: Transparency Justification: Who owns ocean data? Sociology ensures

Dimension	Blue Biotech	Living Resources (Fisheries/Aqua)	R&I (Research & Innovation)	Renewable Energy (Wind/Ocean)	Non-Living Resources (Mining)	Desalination	Maritime Transport	Port Activities	Ship Repair & Shipbuilding	Coastal Tourism	Maritime Defence	Infrastructure & Robotics
A. Understanding	Policy: Nagoya Protocol access & benefit-sharing; t	Policy: fisheries management rules (quotas/catch c	Policy: open science regimes (FAIR data); science-p	Policy: marine renewable energy policy;	Policy: mineral licensing; (where applicable) Interna	Policy: water governance & permitting; brine discha	Policy: IMO convention framework; EU decarboniza	Policy: port governance and port-city policy; hinterl	Policy: ship recycling and hazardous-materials rule	Policy: local coastal tourism strategy; visa/mobility	Policy: EUROSUR situational awareness; CISE cross	Policy: technical standards for offshore/subsea sy
A. Competences	Policy: bio-value communication; systems thinking	Policy: safe seafood communication; ecosystem lit	Policy: blue careers communication; interdisciplina	Policy: public communication of decarb	Policy: resource-limit literacy; thresholds/precautio	Policy: communicating freshwater value; coastal b	Policy: safety communication and reporting; EU ETS	Policy: green port communication; waste-oil and pc	Policy: life-cycle responsibility; participation in Sust	Policy: stewardship communication; tourism ethics	Policy: safety and risk communication; data-sharing	Policy: communicating robotic value/risks; AI gove
B. Digital	Policy: genomics/bioinformatics toolchains; bio-da	Policy: vessel monitoring system (VMS); e-logbook	Policy: modelling/simulation platforms; EMODnet a	Policy: SCADA and load analytics; meto	Policy: remote sensing; geological mapping; digital	Policy: automated operations (process control); eff	Policy: e-navigation; decision support and (future)	Policy: smart logistics; port community systems (P	Policy: digital twin in design/retrofit; 3D CAD; confi	Policy: VR/AR interpretation tools; digital marketing	Policy: surveillance ICT (incl. drones); maritime don	Policy: robot/AUV control; subsea telemetry; opera
C. Sustainability	Policy: bio-active product stewardship; bio-refinery	Policy: selective fishing gear; stock preservation an	Policy: biodiversity restoration R&I; circular innovati	Policy: offshore wind portfolio mix; retro	Policy: carbon sink protection and CCS governance	Policy: water reuse; brine reuse/valorisation; energy	Policy: fuel transition (e-fuels/ammonia/methanol	Policy: sea-level rise adaptation; zero-pollution port	Policy: circular ship dismantling; green shipyards a	Policy: eco-tourism and carrying capacity managen	Policy: greening defence mobility and logistics; dua	Policy: non-toxic materials and pollution preventior
D. Business	Policy: pharma/food value chains; participation in E	Policy: processing chain value; producer networks	Policy: Horizon Europe and related funding; joint R&	Policy: upstream input logistics; industri	Policy: mining support services; strategic metals a	Policy: seawater reverse osmosis (SWRO) referenc	Policy: global trade nodes; carbon pricing and emis	Policy: gateway trade value; accelerators/innovatio	Policy: supply chain integration; green technology n	Policy: experience-economy niches ('new luxury'); lo	Policy: defence R&D and budgets (incl. EU funding	Policy: autonomous systems business models; ser

Policy: the governance / regulatory / institutional frame the competence must operate inside.
Sociology: the key maritime-social-science "problem field" that should shape learning outcomes (labour relations, conflict, justice, values and interests, emotions and traditions, identity, securitization, community impacts, etc.).
Cluster 4: Service & Security layer
Sectors: Coastal Tourism, Maritime Defence, Infrastructure & Robotics

Sectors: Coastal Tourism, Maritime Defence, Infrastructure & Robotics

Dimension	Coastal Tourism	Maritime Defence	Infrastructure & Robotics
A. Understanding	Policy: local coastal tourism strategy, visa/mobility gove	Policy: EUROSUR situational awareness; CISE cross-aut	Policy: technical standards for offshore/subsea systems, permitting & safety cases; seabed/subsea mapping for infrastructure siting Sociology: socio-technical systems; human-automation interaction; deskilling/reskilling; responsibility & auditability in autonomous systems; offshore work cultures
A. Competences	Policy: stewardship communication; tourism ethics and v	Policy: safety and risk communication; data-sharing and I	Policy: communicating robotic value/risks; AI governance and legal compliance Sociology: technology legitimacy; accountability for algorithmic decisions; trust in automation; professional identity shifts for operators
B. Digital	Policy: VR/AR interpretation tools; digital marketing; bo	Policy: surveillance ICT (incl. drones); maritime domain	Policy: robot/AUV control; subsea telemetry; operational data pipelines Sociology: human-in-the-loop design; safety cases for autonomy; operator training regimes; cyber-physical risk governance
C. Sustainability	Policy: eco-tourism and carrying capacity management;	Policy: greening defence mobility and logistics; dual-use	Policy: non-toxic materials and pollution prevention; low-energy operations; life-cycle assessment Sociology: risk governance for offshore operations; OHS in hazardous environments; more-than-human impacts (noise, seabed disturbance)
D. Business	Policy: experience-economy niches ('new luxury'); local	Policy: defence R&D and budgets (incl. EU funding instr	Policy: autonomous systems business models; service ecosystems; standards compliance as market access Sociology: innovation diffusion; platform power; professional communities; regulation as socio-technical boundary

Cluster 2: Hard economy layer

Sectors: Renewable Energy, Non-Living Resources, Desalination

Dimension	Renewable Energy (Wind/Ocean)	Non-Living Resources (Mining)	Desalination
A. Understanding	Policy: marine renewable energy policy; MSP spatial allo	Policy: mineral licensing; (where applicable) International	Policy: water governance & permitting; brine discharge rules; water-energy nexus planning within coastal catchments Sociology: water justice/affordability; techno-politics of infrastructure; public acceptance & risk perception; equity impacts on coastal communities
A. Competences	Policy: public communication of decarbonization benefit	Policy: resource-limit literacy; thresholds/precaution in e	Policy: communicating freshwater value; coastal buffer/setback planning for intake/outfall infrastructure Sociology: public acceptance; perceptions of "manufactured water", equity in tariff design; community concerns about brine impacts
B. Digital	Policy: SCADA and load analytics; metocean and weathe	Policy: remote sensing; geological mapping; digital envir	Policy: automated operations (process control); efficiency and leakage data; cyber-secure OT/IT integration Sociology: socio-technical lock-in; labour deskilling; cybersecurity of critical water infrastructure; transparency in performance claims
C. Sustainability	Policy: offshore wind portfolio mix; retrofits; multi-use c	Policy: carbon sink protection and CCS governance; circu	Policy: water reuse; brine reuse/valorisation; energy-efficiency and pollution-prevention requirements Sociology: water justice; equity impacts of reuse schemes; community risk perception; governance of externalities (brine, energy)
D. Business	Policy: upstream input logistics; industrial clustering; tec	Policy: mining support services; strategic metals and indu	Policy: seawater reverse osmosis (SWRO) reference cases; efficiency and 'smaller footprint' innovation Sociology: infrastructure finance and affordability; public-private governance; inequality in service provision; legitimacy of techno-fixes

Cluster 3: Blue spine layer

Sectors: Maritime Transport, Port Activities, Ship Repair & Shipbuilding

Dimension	Maritime Transport	Port Activities	Ship Repair & Shipbuilding
A. Understanding	Policy: IMO convention framework; EU decarbonization	Policy: port governance and port-city policy; hinterland c	Policy: ship recycling and hazardous-materials rules (Hong Kong Convention and related regimes); green retrofit standards; compliance documentation Sociology: occupational health and safety; subcontracting and precarious work; shipbreaking community impacts; 'just transition' in industrial regions
A. Competences	Policy: safety communication and reporting; EU ETS/Fu	Policy: green port communication; waste-oil and pollution	Policy: life-cycle responsibility; participation in Sustainable Blue Economy Partnership agendas where relevant Sociology: workplace safety culture; skill reproduction in yards; community impacts of ship recycling; governance of subcontracting
B. Digital	Policy: e-navigation; decision support and (future) MASS	Policy: smart logistics; port community systems (PCS) fo	Policy: digital twin in design/retrofit; 3D CAD; configuration management and documentation Sociology: skill regimes and professionalisation; knowledge transfer across subcontractors; cyber-risks in supply chains; human factors in design
C. Sustainability	Policy: fuel transition (e-fuels/ammonia/methanol pathw	Policy: sea-level rise adaptation; zero-pollution port prog	Policy: circular ship dismantling; green shipyards and hazardous-waste control; retrofit capacity Sociology: shipbreaking labour and environmental justice; OHS for toxic materials; skill transitions to green retrofit work
D. Business	Policy: global trade nodes; carbon pricing and emissions	Policy: gateway trade value; accelerators/innovation distr	Policy: supply chain integration; green technology markets (retrofit/efficiency); compliance as competitive advantage Sociology: industrial relations in yards; subcontracting governance; regional development; 'just transition' for shipyard communities

Cluster 1: Bio-economy & knowledge layer

Sectors: Blue Biotech, Living Resources, R&I

Dimension	Blue Biotech	Living Resources (Fisheries/Aqua)	R&I (Research & Innovation)
A. Understanding	Policy: Nagoya Protocol access & benefit-sharing; biores	Policy: fisheries management rules (quotas/catch controls	Policy: open science regimes (FAIR data); science-policy interface for blue governance; research ethics and integrity Sociology: knowledge co-production; participatory governance; boundary work between science and policy; responsible research and innovation
A. Competences	Policy: bio-value communication; systems thinking about	Policy: safe seafood communication; ecosystem literacy (	Policy: blue careers communication; interdisciplinary systems thinking Sociology: epistemic pluralism; transdisciplinarity; reflexivity and ethics in marine social sciences; capacity building
B. Digital	Policy: genomics/bioinformatics toolchains; bio-data gov	Policy: vessel monitoring system (VMS); e-logbooks; ecc	Policy: modelling/simulation platforms; EMODnet access; open repositories and reproducible workflows Sociology: open science norms; reproducibility and trust; citizen science and data ethics; epistemic justice in data infrastructures
C. Sustainability	Policy: bio-active product stewardship; bio-refinery and	Policy: selective fishing gear; stock preservation and eco	Policy: biodiversity restoration R&I; circular innovation portfolios; monitoring and evaluation indicators Sociology: co-production for restoration; socio-ecological experimentation; governance of innovation risks; long-term stewardship institutions
D. Business	Policy: pharma/food value chains; participation in Blue B	Policy: processing chain value; producer networks and m	Policy: Horizon Europe and related funding; joint R&D strategy and partnerships Sociology: political economy of research agendas; knowledge brokerage; inclusivity in innovation systems; evaluation and impact culture

Based on the sources, here is the filled Blue Compe	Kolumna nr 1	Kolumna nr 2	Kolumna nr 3	Kolumna nr 4	Kolumna nr 5	Kolumna nr 6	Kolumna nr 7	Kolumna nr 8	Kolumna nr 9	Kolumna nr 10	Kolumna nr 11	Kolumna nr 12	Kolumna nr 13	Kolumna nr 14
Blue Competence Framework for the Blue Economy														
Dimension (Aspect)	ID	Competence Name	Blue Biotech	Coastal Tourism	Desalination	Infra & Robotics	Living Res.	Non-living Res.	Renewable Energy	Maritime Defence	Maritime Transport	Port Activities	R&I	Ship Repair
A. Understanding the Blue Economy (Skills)	—	Policy & Science Basics	Bio-resource regs	Local strategy use	Water law	Tech standards	Fishery quotas	Mineral law	MRE policy	EUROSUR rules	IMO/FuelEU regs	Port-city policy	Science-policy	MARPOL index
A. Understanding the Blue Economy (Competences	A.1	Ocean Literacy	Bio-value comms	Steward outreach	Fresh water value	Robotic value	Safe seafood info	Resource limits	Green energy ads	Safety comms	Low-carbon ed	Green hub info	Source-to-sea	Blue careers
	A.2	Blue Systems Thinking	Bioprospecting links	LSI mapping	Water cycle cycles	Subsea mapping	Stock-prey links	Sequestration	Grid integration	Border safety	Decarbonization	Hinterland links	Interdisciplinary	Green retrofits
	A.3	Blue Economy Regulations	Nagoya Protocol	Tourism ethics	Buffer planning	AI legal rules	Catch control	Mining threshold	MSP spatial use	CISE data laws	Emission limits	Waste oil rules	SBE Partnership	Scrapping laws
B. Digital & Data Skills (Skills)	—	Software & Cyber Defense	Bio-data use	VR/AR tools	Automated OPS	Robot control	VMS tracking	Remote sensing	Load analytics	Surveillance ICT	MASS navigation	Smart logistics	Digital Twin use	Simulation software
B. Digital & Data Skills (Competences)	B.1	Data & Digital Proficiency	Genomic data	Tourism GIS	Efficiency data	AUV telemetry	Ecosystem models	Geology mapping	Weather models	Radar analysis	IT route optimization	Blockchain tracking	FAIR data access	3D design
	B.2	Digital Communication	Cross-BSR nets	Marina networks	Stakeholder info	IoT sensors	Producer nets	Global trade info	Stakeholder MSP	Interoperable CISE	Logistics sharing	Port accelerator	Open Sci platforms	System responsibles
	B.3	Cybersecurity & Maritime Safety	Bio-data security	Booking security	Process integrity	Cable protection	Safe work tech	Extractive safety	Maintenance safety	CISE protection	Navigation safety	Port NIS2 plans	Vulnerability info	Risk analysis
	B.4	Open Science & Data Sharing	Bio-Biotech Forum	BASTIS portal	R&D results	Subsea data	Citizen science	EMODnet access	SRIA agendas	Multi-use data	MRV fuel logs	Logistics data	Copernicus access	Shared R&D
C. Sustainability & Resilience (Skills)	—	Green Design & Resource Planning	Bioremediation	Smaller footprint	Solar desalination	Durable materials	Gear innovation	Aggregate limits	Floating structures	Military mobility	Clean propulsions	OPS shore power	Knowledge transfer	Eco-ship builds
C. Sustainability & Resilience (Competences)	C.1	Sustainable Resource Management	Bioactive use	Water reuse	Low-energy water	Subsea cables	Stock preservation	Carbon sinks	Carbon mitigation	Resource security	Emission reduction	Waste handling	Ecosystem accounts	Hazardous waste
	C.2	Circular Economy Principles	Bio-refinery	Waste management	Brine reuse	Recycling cables	By-product use	Sea-salt mining	Multi-use wind	Civilian synergy	Low-impact hulls	Zero-pollution port	Circular R&I	Ship dismantling
	C.3	Climate Adaptation & Coastal Resilience	Habitat care	Flood protection	Drought response	Hab monitoring	Resilient fish	Shore restoration	Offshore wind mix	Risk assessment	Fuel transition	Port sea-level rise	Hazard monitoring	Retrofitting tech
	C.4	Ecosystem-based Management (EBM)	Regenerative algae	Eco-tourism	Environ assessment	Non-toxic robotics	Selective fishing	Benthic health	MPA synergy	Environ monitor	Noise pollution	Biodiversity restoration	Holistic care	Green shipyards
D. Business Innovation & Governance (Skills)	—	Value Mapping & Financial Tracking	Patent tracking	GVA analysis	Method analysis	R&D investment	Chain profitability	Market potential	LCOE analysis	Budget planning	Fuel cost management	Port logistics value	GVA monitoring	Product innovation
D. Business Innovation & Governance (Competence	D.1	Value Chain Thinking	Pharma/Food uses	Local job creation	SWRO reference	Service ecosystem	Processing chain	Mining support	Upstream input	Military mobility	Logistics nodes	Gateway trade	Knowledge triangle	Supply chain integration
	D.2	Leadership & Blue Innovation	SME accelerators	"New Luxury" niches	R&D methods	Autonomous system	New species dev	High-tech metals	Tech leadership	Strategic autonomy	Zero-emission ship	Accelerator schemes	KIC integration	Green technology
	D.3	Blue Finance & Investment	BlueInvest grants	EU Recovery Fund	PPP projects	High cost R&D	Incentives to SMEs	Refund guarantees	Taxonomy finance	Defence spending	Emission trading	Smart infrastructure	R&I funding	Risk tools (Fintech)
	D.4	Ethical & Participatory Governance	Blue Bio Forum	Stakeholder benefit	Strategic buffering	Comm. building	Fairness in basins	Geopolitical alliance	MSP co-usage	Civil-military sync	Social inclusion	Port Commissions	Science-policy Society	Joint R&D strategy

	Cluster 4: Tourism, Defence & Robotics (The "Service & Security" Layer). Focus: Experience, Safety, and High-Tech Intervention.			Cluster 2: Energy, Resources & Desalination (The "Hard" Economy). Focus: Decarbonization, Engineering, and Spatial Management			Cluster 3: Transport, Ports & Infrastructure (The "Blue Spine"). Focus: Logistics, Digital Twins, and Global Trade.			Cluster 1: Living Resources & Biotechnology (The "Bio-Economy"). Focus: Stewardship, Circularity, and High-Tech Value Creation: Focus: Stewardship, Circ			
Dimension	Coastal Tourism	Maritime Defence	Infrastructure & Robotics	Renewable Energy (Wind/Ocean)	Non-Living Resources (Mining)	Desalination	Maritime Transport	Port Activities	Ship Repair & Shipbuilding	Blue Biotech	Living Resources (Fisheries/Aqua)	R&I (Research & Innovation)	Kolumna nr 1
A. Understanding	Regs: Local strategy / VisasA.2: Land-Sea Interacti	Regs: EUROSUR / CISE rulesA.2: Border safety	Regs: Tech standardsA.2: Subsea mapping	Regs: MRE policy / Spatial useA.2: Grid integration	Regs: Mineral law / Mining codeA.2: Sequestration	Regs: Water law / Brine rulesA.2: Water cycle cycle	Regs: IMO / FuelEU MaritimeA.2: Decarbonization	Regs: Port-city policyA.2: Hinterland links	Regs: Scrapping laws / HK Conv.A.2: Green retrofit	Regs: Nagoya ProtocolPolicy: Bio-resource regs	Regs: Catch control / QuotasPolicy: Fishery quotas	Regs: Open Science (FAIR)Policy: Science-policy i	
A. Competences	A.1: Stewardship outreachA.3: Tourism ethics	A.1: Safety commsA.3: CISE data laws	A.1: Robotic valueA.3: AI legal rules	A.1: Green energy advocacyA.3: MSP spatial de-co	A.1: Resource limits awarenessA.3: Mining threshol	A.1: Fresh water valueA.3: Buffer planning	A.1: Safety communicationA.3: Emission limits (ET	A.1: Green hub infoA.3: Waste oil rules	A.1: Lifecycle awarenessA.3: SBE Partnership	A.1: Bio-value communicationA.2: Bioprospecting l	A.1: Safe seafood infoA.2: Stock-prey links	A.1: Blue careers advocacyA.2: Interdisciplinary thi	
B. Digital	Skills: VR/AR toolsB.3: Booking security	Skills: Surveillance ICT / DronesB.3: CISE protectic	Skills: Robot control / AUVsB.1: AUV telemetry	Skills: Load analytics / SCADAB.1: Weather models	Skills: Remote sensingB.1: Geology mapping	Skills: Automated OPSB.1: Efficiency data	Skills: MASS navigation / AIB.1: IT route optimizatio	Skills: Smart logistics / PCSB.2: Logistics sharing	Skills: Digital Twin (Design)B.1: 3D design / CAD	Skills: Bio-data / GenomicsB.1: Genomic data profi	Skills: VMS tracking / e-LogbooksB.1: Ecosystem m	Skills: Simulation softwareB.4: EMODnet access	
C. Sustainability	C.4: Eco-tourism / Carrying cap.C.1: Water/Waste r	C.1: Military mobility (Green)D.4: Civil-military sync	C.4: Non-toxic roboticsC.1: Low-energy water ops	C.3: Offshore wind mix / RetrofitC.2: Multi-use wind	C.1: Carbon sinks / CCSC.2: Sea-salt mining (Brine	C.1: Water reuseC.2: Brine reuse	C.3: Fuel transition (e-fuels)C.4: Noise pollution red	C.3: Sea-level rise adaptationC.2: Zero-pollution po	C.2: Ship dismantling / CircularityC.4: Green shipya	C.1: Bioactive useC.2: Bio-refinery (Zero waste)	C.4: Selective fishing gearC.1: Stock preservation	C.4: Biodiversity restorationC.2: Circular R&I	
D. Business	D.2: "New Luxury" nichesD.1: Local job creation	D.3: Defence spending (EDF)D.4: Geopolitical allia	D.2: Autonomous systemsD.1: Service ecosystem	D.1: Upstream input logisticsD.2: Tech leadership	D.1: Mining support servicesD.2: High-tech metals	D.1: SWRO reference casesD.2: Smaller footprint t	D.1: Global trade nodesD.3: Emission trading (ETS	D.1: Gateway trade valueD.2: Accelerator schemes	D.1: Supply chain integrationD.2: Green technology	D.1: Pharma/Food value chainsD.3: Blue Bio Forur	D.1: Processing chain valueD.4: Producer networks	D.3: Horizon Europe fundingD.4: Joint R&D Strateg	

Here is the plain text explanation of the four strategic clusters. This breakdown explains why these specific sectors are grouped together and how they function as a coherent unit for University of Szczecin:

Cluster 1: Living Resources & Biotechnology (The "Bio-Economy")
The Focus: This cluster is entirely centered on biological life—from the microscopic genetic material of algae to the management of massive fish stocks.

The Logic:
From Extraction to Stewardship: Historically, fisheries were about "hunting." The new logic, driven by the Ocean Pact, is about stewardship (caring for the stock) and cultivation (aquaculture).
High-Tech Value: We group Fisheries with Biotech and R&I because the future value of the ocean isn't just in selling fish fillets, it's in discovering new pharmaceuticals, cosmetics, and biofuels from marine organisms.
Sociological Angle: This is where the concept of the "Ocean Steward" is born. The key skill here isn't just catching fish, but understanding the ecosystem so that the community can survive for generations (intergenerational justice).

Cluster 2: Energy, Resources & Desalination (The "Hard" Economy)
The Focus: This cluster deals with industrial production and engineering within the marine space. It covers offshore wind farms, seabed mining, and freshwater production.

The Logic:
Space is the Resource: Unlike fishing (where fish move), these industries need fixed space. The central challenge here is Maritime Spatial Planning (MSP)—figuring out how to put a wind farm next to a shipping lane without causing conflict.
Decarbonization: This cluster is the "engine room" of the Green Deal. The logic is purely technical and industrial: how to generate maximum power with minimum environmental harm.
Sociological Angle: This is the zone of highest conflict (e.g., fishermen protesting wind farms). The sociological competence needed here is conflict mediation and obtaining a "Social License to Operate."

Cluster 3: Transport, Ports & Infrastructure (The "Blue Spine")
The Focus: This cluster focuses on logistics, movement, and trade. It connects the ocean to the land (Hinterland) via ports, ships, and repair yards.

The Logic:
Efficiency & Data: The global economy relies on this spine. The modern logic is Digitization—using "Digital Twins" to track every container and emission molecule to optimize efficiency.
The Green Transition: Shipping is dirty; the goal here is to retrofit the entire fleet for green fuels (ammonia/hydrogen) and circularity (recycling old ships).
Sociological Angle: This cluster addresses the "Port-City" relationship. The goal is Re-maritimisation—ensuring the port city (like Szczecin) doesn't turn its back on the port, but integrates it culturally and economically.

Cluster 4: Tourism, Defence & Robotics (The "Service & Security" Layer)
The Focus: This cluster encompasses the immaterial (tourism/experience) and the protective (defence/surveillance), enabled by advanced robotics.

The Logic:
Access without Extraction: Tourism sells the experience of the sea; Defence ensures the safety of the sea. Neither extracts physical resources, but both require deep knowledge of the environment.
The Eyes of the Ocean: We include Robotics here because drones and sensors are the "eyes" that allow tourists to see reefs (VR/AR) and navies to see threats (surveillance).
Sociological Angle: This is about Ocean Literacy and Security. It transforms the public from passive observers into active "Ocean Citizens" who understand and value the marine environment.